

Arithmetic Structures of Local Gromov–Witten Theory for Non-nef Calabi–Yau Curves

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Abstract

We study the arithmetic structure of the antidiagonal equivariant genus-zero local Gromov–Witten theory of

$$X_k = \text{Tot}(\mathcal{O}_{\mathbb{P}^1}(k) \oplus \mathcal{O}_{\mathbb{P}^1}(-2-k)), \quad k \geq 1.$$

Starting from the closed local mirror-symmetry formula of Forbes–Jinzenji, we rewrite the degree- d Gromov–Witten coefficient in binomial form and analyze its multiple-cover transform by prime-power congruences. General integrality of Gopakumar–Vafa invariants for toric Calabi–Yau threefolds is known from work of Konishi and Peng; our aim is instead to exhibit an explicit arithmetic mechanism for this particular non-nef local-curve family. We first prove a classification theorem for the two-parameter binomial family

$$A_{R,\varepsilon}(n) = (-1)^{\varepsilon(n-1)} \binom{Rn-1}{n-1}, \quad R \geq 2, \quad \varepsilon \in \{0, 1\},$$

showing that the all-prime cubic prime-step congruence holds precisely when R is a quadratic residue modulo 12 and the parity character is the one forced by R modulo 4. The local Calabi–Yau specialization $R = (k+1)^2$, $\varepsilon \equiv k \pmod{2}$ therefore falls automatically into the classified range. We then derive the corresponding d^3 -divisibility of the Möbius numerator. For odd primes the argument is controlled by refined Jacobsthal–Kazandzidis congruences. At the exceptional prime 2 we use Gessel’s sharp binomial congruence; its exceptional sign agrees exactly with the parity character arising from the local Calabi–Yau parameter. We also give an explicit Lagrange–Bürmann realization of the unsigned binomial sequence from the rational function $z/(1-z)$ and explain its relation to the framing framework for 2-functions, improved integrality, and 3-sequences without relying on a convention-dependent identification.

Finally, we prove that the BPS sequence has the same leading large-degree asymptotics as the Gromov–Witten sequence, with exponentially smaller Möbius corrections. For $\mathcal{O}(1) \oplus \mathcal{O}(-3) \rightarrow \mathbb{P}^1$ we also record the exact degree recursion, hypergeometric form, and the first BPS values.

Keywords: local Gromov–Witten invariants, BPS invariants, non-nef toric Calabi–Yau threefolds, supercongruences, Möbius inversion, framing
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1. Introduction

Consider the toric Calabi–Yau threefold

$$X_k = \text{Tot}(\mathcal{O}_{\mathbb{P}^1}(k) \oplus \mathcal{O}_{\mathbb{P}^1}(-2-k)), \quad k \geq 1.$$

Forbes and Jinzenji developed equivariant local mirror symmetry for this non-nef family by means of equivariant I -functions and Birkhoff factorization [1, 2]. Under the antidiagonal specialization their genus-zero coefficient can be written as

$$N_d^{(k)} = \frac{(-1)^{(d-1)k}}{d^3} \binom{(k+1)^2 d - 1}{d-1}. \quad (1)$$

Formula (1) is an established local mirror-symmetry input. We stress also that integrality of Gopakumar–Vafa invariants for general toric Calabi–Yau threefolds was proved by Konishi [3] and, by a different method, by Peng [4]. Thus the novelty sought here is not a new proof of the general toric Gopakumar–Vafa conjecture, but an explicit prime-by-prime supercongruence for the closed binomial sequence attached to X_k , together with a direct framing interpretation. Very recently Guo and Zhang studied the same family X_k by the topological vertex and proved closed formulas for leading Gopakumar–Vafa invariants [5]. Their results make it especially important to distinguish the present genus-zero Möbius/supercongruence problem from top-genus or leading-genus calculations.

Set

$$R = (k+1)^2, \quad A_k(n) = (-1)^{(n-1)k} \binom{Rn-1}{n-1}.$$

The genus-zero multiple-cover relation suggests

$$n_d^{(k)} = \frac{1}{d^3} \sum_{m|d} \mu(m) A_k(d/m). \quad (2)$$

37 Thus the BPS integrality problem is equivalent to a cubic divisibility statement
 38 for a Möbius transform.

39 Our main theorem is the following.

40 **Theorem 1.1** (Cubic divisibility and BPS integrality). *For all integers*
 41 *$k, d \geq 1$,*

$$d^3 \mid \sum_{m \mid d} \mu(m) (-1)^{(d/m-1)k} \binom{(k+1)^2 d/m - 1}{d/m - 1}. \quad (3)$$

42 *Consequently,*

$$n_d^{(k)} = \frac{1}{d^3} \sum_{m \mid d} \mu(m) (-1)^{(d/m-1)k} \binom{(k+1)^2 d/m - 1}{d/m - 1} \in \mathbb{Z}.$$

43 The proof is local at primes. For $p \geq 3$ we use the refined Jacobsthal–
 44 Kazandzidis congruence. The prime 2 is the interesting part. Gessel’s sharp
 45 form of the binomial congruence contains one exceptional case with a minus
 46 sign. For

$$a = 2Rn, \quad b = 2n,$$

47 that exceptional case occurs exactly when R is even and n is odd. Since
 48 $R = (k+1)^2$, this is exactly the case k odd and n odd, and the exceptional
 49 minus sign is canceled by the factor $(-1)^{(n-1)k}$ in $A_k(n)$. This gives an
 50 all-prime prime-step congruence with cubic precision.

51 A second observation is that the unsigned sequence

$$\binom{Rn - 1}{n - 1}$$

52 admits a direct Lagrange–Bürmann realization from the rational function
 53 $z/(1-z)$ with parameter $R-1$. This realization has the same formal shape
 54 as the framing constructions appearing in the theory of 2-functions. Because
 55 framing conventions in the literature differ by choices of sign, coordinate, and
 56 normalization, we keep the coefficient identity independent of that terminology
 57 and compare it only at the level justified by the present formulas. Müller’s
 58 work [6] provides the relevant arithmetic context, relating framings of rational
 59 2-functions to improved 3-integrality.

60 For the basic case

$$X_1 = \text{Tot}(\mathcal{O}(1) \oplus \mathcal{O}(-3))$$

61 we obtain

$$N_d = \frac{(-1)^{d+1}}{4d^3} \binom{4d}{d},$$

62 and the first BPS numbers are

$$1, -1, 2, -7, 31, -156, \dots$$

63 *Geometry–physics viewpoint..* The antidiagonal specialization is not merely
64 a convenient combinatorial normalization: it is the equivariantly Calabi–
65 Yau choice in the local curve model and is precisely the specialization for
66 which the mirror data simplify to a one-parameter structure. The resulting
67 rational Gromov–Witten coefficients encode multiple covers, while the Möbius
68 transform isolates the associated genus-zero BPS coefficients. Our arithmetic
69 question is therefore geometric in origin: which prime-power congruences
70 force the multiple-cover transform of this particular local mirror sequence
71 to be integral? The answer below shows that the low-prime congruences
72 remember the square parameter $R = (k+1)^2$ and its parity character.

73 *Relation to previous work*

74 The local curve theory sits at the intersection of several established
75 developments. The Gopakumar–Vafa multiple-cover structure originates in
76 [7, 8]; the toric calculation is governed by the topological vertex [9, 10], while
77 the local curve formalism is developed in [11]. General toric GV integrality
78 is already known [3, 4]. On the arithmetic side, our proof uses the binomial
79 supercongruences of Kazandzidis, Gessel, and Robert–Zuber [12, 13, 14]. The
80 connection with p -adic integrality in mirror symmetry goes back, in a different
81 form, to Kontsevich–Schwarz–Vologodsky [15]. Framing and 2-functions were
82 developed by Schwarz–Vologodsky–Walcher [16], and Müller proved improved
83 3-integrality results for framings of rational 2-functions [6]. Zhu’s quantum
84 2-function formalism gives a related topological-string perspective [17].

85 *Organization..* The paper is organized into five sections. Section 2 collects the
86 geometric input from antidiagonal equivariant local mirror symmetry and for-
87 mulates the multiple-cover transform. Section 3 contains the arithmetic core:
88 the classification theorem, the exceptional two-adic analysis, and the cubic
89 BPS divisibility theorem. Section 4 develops the Lagrange–Bürmann realiza-
90 tion, the model $\mathcal{O}(1) \oplus \mathcal{O}(-3)$, and the large-degree GW/BPS asymptotics.
91 Section 5 concludes with geometric and arithmetic perspectives.

92 *Main contributions..* The paper has four closely related outputs. First, it gives
 93 a direct prime-by-prime proof of cubic divisibility for the Möbius numerator
 94 of the antidiagonal local curve sequence. Second, it proves a classification
 95 of the two-parameter binomial sequences $A_{R,\epsilon}$ satisfying the same all-prime
 96 cubic prime-step congruence; the local Calabi–Yau values $R = (k + 1)^2$ form
 97 a canonical square subclass. Third, the proof identifies a specific two-adic
 98 cancellation: Gessel’s exceptional sign is exactly balanced by the parity
 99 character supplied by the local curve parameter. Finally, the BPS transform
 100 is shown to preserve the leading exponential growth of the Gromov–Witten
 101 coefficients, with exponentially smaller proper-divisor corrections.

102 2. Local Calabi–Yau geometry, mirror symmetry, and multiple 103 covers

104 The total space

$$X_k = \text{Tot}(\mathcal{O}_{\mathbb{P}^1}(k) \oplus \mathcal{O}_{\mathbb{P}^1}(-2 - k))$$

105 is Calabi–Yau because

$$K_{\mathbb{P}^1} \otimes \det(\mathcal{O}(k) \oplus \mathcal{O}(-2 - k))^{-1} \simeq \mathcal{O}_{\mathbb{P}^1}.$$

106 Although toric, it is non-nef for $k > 0$. Forbes–Jinzenji developed the
 107 equivariant local mirror theory of this family by equivariant I -functions and
 108 Birkhoff factorization, and for the antidiagonal action obtained closed mirror-
 109 map and Yukawa data [1, 2]. We use the resulting coefficient form below as
 110 the geometric input; the new arguments in this paper begin with its arithmetic
 111 analysis.

112 **Proposition 2.1** (Antidiagonal local mirror coefficient). *Let $R = (k + 1)^2$.
 113 In the antidiagonal equivariant genus-zero theory,*

$$N_d^{(k)} = \frac{(-1)^{(d-1)k}}{d^3} \binom{Rd - 1}{d - 1}.$$

114 The remainder of the paper concerns the arithmetic consequences of this
 115 formula.

116 *2.1. Multiple covers and Möbius inversion*

117 Assume the genus-zero multiple-cover decomposition

$$N_d^{(k)} = \sum_{m|d} \frac{n_{d/m}^{(k)}}{m^3}.$$

118 Möbius inversion yields

$$n_d^{(k)} = \frac{1}{d^3} \sum_{m|d} \mu(m) A_k(d/m), \quad A_k(n) = (-1)^{(n-1)k} \binom{Rn-1}{n-1}.$$

119 Put

$$B_k(d) := \sum_{m|d} \mu(m) A_k(d/m).$$

120 **Lemma 2.2** (Möbius pairing). *If $p^\alpha \| d$ and $d = p^\alpha q$ with $(p, q) = 1$, then*

$$B_k(d) = \sum_{u|q} \mu(u) [A_k(p^\alpha q/u) - A_k(p^{\alpha-1} q/u)].$$

121 *Proof.* In the Möbius sum only divisors with p -adic exponent 0 or 1 survive.

122 Pair $m = u$ and $m = pu$ and use $\mu(pu) = -\mu(u)$. $\square$

123 Thus it suffices to prove the following prime-step statement.

124 For the arithmetic argument it is useful to separate the special Calabi–Yau
125 square parameter from the underlying binomial phenomenon. For an integer
126 $R \geq 2$ and $\varepsilon \in \{0, 1\}$ define

$$A_{R,\varepsilon}(n) = (-1)^{\varepsilon(n-1)} \binom{Rn-1}{n-1}.$$

127 **Theorem 2.3** (Classification of cubic binomial prime-step congruences). *The*
128 *following are equivalent.*

129 1. *For every prime p and every $n \geq 1$,*

$$A_{R,\varepsilon}(pn) \equiv A_{R,\varepsilon}(n) \pmod{p^{3(\text{ord}_p(n)+1)}}.$$

130 2. One has

$$R \equiv 0, 1, 4, 9 \pmod{12},$$

131 and, more precisely,

$$\varepsilon = \begin{cases} 0, & R \equiv 1 \pmod{4}, \\ 1, & R \equiv 0 \pmod{4}. \end{cases}$$

132 Equivalently, R is a square modulo 12 and the parity character is uniquely
133 determined by the residue of R modulo 4.

134 **Corollary 2.4** (Local Calabi–Yau specialization). *For*

$$R = (k + 1)^2, \quad \varepsilon \equiv k \pmod{2},$$

135 the conditions of Theorem 2.3 are automatically satisfied. Consequently, for
136 every prime p and $s = \text{ord}_p(n)$,

$$A_k(pn) \equiv A_k(n) \pmod{p^{3(s+1)}}.$$

137 3. Cubic supercongruences and BPS integrality

138 We shall repeatedly use

$$A_{R,\varepsilon}(n) = \frac{(-1)^{\varepsilon(n-1)}}{R} \binom{Rn}{n}.$$

139 3.1. Necessity of the residue conditions

140 Assume the prime-step congruence in Theorem 2.3. Taking $p = 2$ and
141 $n = 1$ gives

$$(-1)^\varepsilon(2R - 1) \equiv 1 \pmod{8}.$$

142 If $\varepsilon = 0$, this is equivalent to $R \equiv 1 \pmod{4}$; if $\varepsilon = 1$, it is equivalent to
143 $R \equiv 0 \pmod{4}$.

144 Next take $p = 3$ and $n = 1$. Since the sign is automatically positive,

$$\binom{3R - 1}{2} \equiv 1 \pmod{27}.$$

145 A direct simplification gives

$$\binom{3R - 1}{2} - 1 = \frac{9R(R - 1)}{2},$$

146 hence

$$3 \mid R(R-1).$$

147 Therefore $R \equiv 0$ or $1 \pmod{3}$. Combining the conditions modulo 3 and
148 modulo 4 yields

$$R \equiv 0, 1, 4, 9 \pmod{12}.$$

149 3.2. A uniform binomial congruence

150 All prime cases will be deduced from a single theorem of Gessel [13]. We
151 record precisely the form used below.

152 **Lemma 3.1** (Gessel). *Let a, b be nonnegative integers divisible by a prime p ,
153 with $0 \leq b \leq a$. Put*

$$\alpha = \text{ord}_p(a), \quad \beta = \text{ord}_p(b), \quad \gamma = \text{ord}_p(a-b), \quad \delta = \text{ord}_p\left(\frac{a/p}{b/p}\right),$$

154 and let

$$\mu_p = \begin{cases} 2, & p = 2, \\ 1, & p = 3, \\ 0, & p > 3. \end{cases}$$

155 Except when

$$p = 2, \quad b \equiv a - b \equiv 2 \pmod{4},$$

156 one has

$$\binom{a}{b} \equiv \binom{a/p}{b/p} \pmod{p^{\alpha+\beta+\gamma+\delta-\mu_p}}.$$

157 In the exceptional case the same congruence holds with $-\binom{a/p}{b/p}$ in place of
158 $\binom{a/p}{b/p}$.

159 3.3. Odd primes

160 **Lemma 3.2.** *Under the residue hypotheses of Theorem 2.3, the prime-step
161 congruence holds for every odd prime p .*

162 *Proof.* Let $s = \text{ord}_p(n)$ and apply Lemma 3.1 with

$$a = pRn, \quad b = pn.$$

163 Write

$$r = \text{ord}_p(R), \quad t = \text{ord}_p(R-1), \quad \delta = \text{ord}_p\left(\frac{Rn}{n}\right).$$

164 Then

$$\alpha = 1 + r + s, \quad \beta = 1 + s, \quad \gamma = 1 + t + s.$$

165 For odd p there is no exceptional sign, and Gessel gives

$$\text{ord}_p \left(\binom{pRn}{pn} - \binom{Rn}{n} \right) \geq 3 + r + t + 3s + \delta - \mu_p.$$

166 After division by R , the normalized difference therefore has valuation at least

$$3 + t + 3s + \delta - \mu_p.$$

167 If $p > 3$, then $\mu_p = 0$, so this is at least $3(s + 1)$. If $p = 3$, then $\mu_p = 1$.

168 The classification hypotheses imply $R \equiv 0$ or $1 \pmod{3}$. In the first case,

$$\delta = \text{ord}_3 \left(\binom{Rn}{n} \right) \geq \text{ord}_3(R) \geq 1$$

169 because

$$\binom{Rn}{n} = R \binom{Rn-1}{n-1}.$$

170 In the second case $t = \text{ord}_3(R-1) \geq 1$. Hence again the normalized valuation
171 is at least $3(s + 1)$.

172 Finally, for every odd prime,

$$(-1)^{\varepsilon(pn-1)} = (-1)^{\varepsilon(n-1)},$$

173 so the parity character does not alter the congruence. $\square$

174 3.4. The prime two

175 We now apply the same Gessel congruence at the exceptional prime 2.

176 **Lemma 3.3.** *Under the residue and parity hypotheses of Theorem 2.3, the*
177 *prime-step congruence holds for $p = 2$.*

178 *Proof.* Let $s = \text{ord}_2(n)$ and set

$$C_n = \binom{Rn}{n}.$$

179 Apply Lemma 3.1 with

$$a = 2Rn, \quad b = 2n.$$

180 Write

$$r = \text{ord}_2(R), \quad t = \text{ord}_2(R-1), \quad \delta = \text{ord}_2(C_n).$$

181 Then

$$\alpha = 1 + r + s, \quad \beta = 1 + s, \quad \gamma = 1 + t + s.$$

182 Hence the modulus supplied by Gessel has exponent

$$\alpha + \beta + \gamma + \delta - 2 = 1 + r + t + 3s + \delta.$$

183 Using $A_{R,\varepsilon}(n) = (-1)^{\varepsilon(n-1)} R^{-1} \binom{Rn}{n}$ and dividing by R , the exponent becomes

$$1 + t + 3s + \delta.$$

184 We claim that

$$t + \delta \geq 2.$$

185 If $R \equiv 1 \pmod{4}$, then $t = \text{ord}_2(R-1) \geq 2$. If $R \equiv 0 \pmod{4}$, then $r \geq 2$,
186 and

$$C_n = R \binom{Rn-1}{n-1}$$

187 implies $\delta \geq r \geq 2$. Thus the normalized modulus has exponent at least

$$1 + 3s + 2 = 3(s+1).$$

188 It remains to check the exceptional sign. The exceptional condition is

$$2n \equiv 2 \pmod{4}, \quad 2(R-1)n \equiv 2 \pmod{4},$$

189 which is equivalent to n odd and R even. Under the hypotheses of Theorem 2.3,
190 R even means $R \equiv 0 \pmod{4}$ and therefore $\varepsilon = 1$. Gessel's exceptional minus
191 sign is then canceled by

$$\frac{(-1)^{\varepsilon(2n-1)}}{(-1)^{\varepsilon(n-1)}} = -1.$$

192 Outside the exceptional case Gessel has the positive sign. If $\varepsilon = 0$ there is
193 nothing to check. If $\varepsilon = 1$, then R is even and nonexceptionality forces n
194 even, so both parity factors are equal. Hence in all cases

$$A_{R,\varepsilon}(2n) \equiv A_{R,\varepsilon}(n) \pmod{2^{3(s+1)}}.$$

195

□

196 *Proof of Theorem 2.3.* The necessity was proved above. For sufficiency com-
 197 bine Lemmas 3.2 and 3.3. $\square$

198 *Proof of Corollary 2.4.* Every integer square is congruent to 0, 1, 4, or 9 mod-
 199 ulo 12. Moreover $(k+1)^2 \equiv 1 \pmod{4}$ when k is even and $(k+1)^2 \equiv 0$
 200 $\pmod{4}$ when k is odd. Thus the required parity parameter is exactly $\varepsilon \equiv k$
 201 $\pmod{2}$. $\square$

202 *3.5. From prime-step congruences to cubic BPS divisibility*

203 *Proof of Theorem 1.1.* Fix $p^\alpha \parallel d$ and write $d = p^\alpha q$ with $(p, q) = 1$. By
 204 Lemma 2.2,

$$B_k(d) = \sum_{u|q} \mu(u) [A_k(p^\alpha q/u) - A_k(p^{\alpha-1} q/u)].$$

205 For

$$n = p^{\alpha-1} q/u$$

206 we have $\text{ord}_p(n) = \alpha - 1$. Corollary 2.4 therefore implies that each bracket is
 207 divisible by $p^{3\alpha}$. Hence $p^{3\alpha} \mid B_k(d)$. This holds for every prime power $p^\alpha \parallel d$,
 208 and the corresponding prime powers are pairwise coprime. Therefore

$$d^3 = \prod_{p^\alpha \parallel d} p^{3\alpha} \mid B_k(d).$$

209 The integrality of $n_d^{(k)} = B_k(d)/d^3$ follows. $\square$

210 *Remark 3.4* (Why the square parameter matters). For $R = (k+1)^2$ one has

$$R \equiv 0, 1 \pmod{3}.$$

211 At the prime 2 the relevant statement is not $R \equiv 0, 1 \pmod{8}$: squares
 212 modulo 8 may also be congruent to 4. Rather,

$$k \text{ even} \implies R \equiv 1 \pmod{8} \implies 8 \mid R - 1,$$

213 whereas

$$k \text{ odd} \implies 4 \mid R.$$

214 These are exactly the two sources of the additional 2-adic divisibility used
 215 above. The remaining exceptional sign is precisely the character $(-1)^{(n-1)k}$.

216 4. Lagrange realization, examples, and large-degree asymptotics

217 We first establish a formal coefficient identity without imposing any
 218 external framing convention. This avoids a normalization ambiguity: in
 219 the 2-function literature, framing may be written using different choices of
 220 coordinate, sign, and primitive. The arithmetic results of the preceding
 221 sections do not depend on this terminology.

222 Let

$$V(z) = \frac{z}{1-z}.$$

223 Then

$$\int V(z) = -\log(1-z),$$

224 where $\int(z^n) = z^n/n$. Let $\nu \in \mathbb{Z}_{\geq 0}$ and define a new variable

$$w = z(1-z)^\nu.$$

225 Let $z = z(w)$ be the formal inverse and define

$$\mathcal{L}_\nu[V](w) := \frac{z(w)}{1 - (\nu+1)z(w)}.$$

226 **Proposition 4.1** (Lagrange coefficient formula). *For $\nu \geq 0$,*

$$\mathcal{L}_\nu[V](w) = \sum_{n \geq 1} \binom{(\nu+1)n-1}{n-1} w^n.$$

227 *Proof.* From $w = z(1-z)^\nu$ we have

$$z = w(1-z)^{-\nu}.$$

228 Apply Lagrange–Bürmann inversion to

$$f(z) = \frac{z}{1 - (\nu+1)z}.$$

229 Since

$$f'(z) = \frac{1}{(1 - (\nu+1)z)^2},$$

230

$$[w^n]f(z(w)) = \frac{1}{n}[z^{n-1}] \frac{(1-z)^{-\nu n}}{(1 - (\nu+1)z)^2}.$$

231 The coefficient evaluates to

$$\binom{(\nu+1)n-1}{n-1},$$

232 which proves the claim. $\square$

233 Taking

$$\nu = R - 1 = (k + 1)^2 - 1 = k(k + 2)$$

234 gives precisely the unsigned sequence underlying $A_k(n)$. Thus Proposition 4.1
235 supplies an intrinsic Lagrange realization of the binomial coefficients appearing
236 in the local curve theory.

237 This formula is closely related to the framing formalism for rational 2-
238 functions studied by Schwarz–Vologodsky–Walcher and Müller [16, 6]. We
239 deliberately do not identify $\mathcal{L}_\nu$ with a named positive or negative framing
240 operation without fixing all of their coordinate and sign conventions. What
241 is used in this paper is only the explicit coefficient identity above. Müller’s
242 theorem, that framings of rational 2-sequences exhibit improved 3-integrality,
243 therefore serves as conceptual arithmetic context rather than as an input to
244 our proof.

245 *4.1. The model geometry $\mathcal{O}(1) \oplus \mathcal{O}(-3)$*

246 For $k = 1$ one has $R = 4$ and

$$N_d = \frac{(-1)^{d+1}}{d^3} \binom{4d-1}{d-1} = \frac{(-1)^{d+1}}{4d^3} \binom{4d}{d}. \quad (4)$$

247 *4.2. Exact recursion*

248 **Proposition 4.2.** *The sequence is determined by $N_1 = 1$ and*

$$N_{d+1} = -\frac{4d^3(4d+1)(4d+2)(4d+3)}{3(d+1)^4(3d+1)(3d+2)} N_d.$$

249 *Proof.* Take the ratio of consecutive terms in (4). $\square$

250 *4.3. Hypergeometric form*

251 Let

$$H(q) = \sum_{d \geq 0} \binom{4d}{d} q^d.$$

252 Then

$$H(q) = {}_3F_2 \left(\frac{1}{4}, \frac{1}{2}, \frac{3}{4}; \frac{1}{3}, \frac{2}{3}; \frac{256}{27} q \right).$$

253 If

$$F_0(q) = \sum_{d \geq 1} N_d q^d, \quad \Theta = q \frac{d}{dq},$$

254 then

$$\Theta^3 F_0(q) = \frac{1}{4} \left[1 - {}_3F_2 \left(\frac{1}{4}, \frac{1}{2}, \frac{3}{4}; \frac{1}{3}, \frac{2}{3}; -\frac{256}{27} q \right) \right].$$

255 *4.4. Low-degree GW and BPS data*

d	N_d	n_d
1	1	1
2	$-\frac{7}{8}$	-1
3	$\frac{55}{27}$	2
4	$-\frac{455}{64}$	-7
5	$\frac{3876}{125}$	31
6	$-\frac{33649}{216}$	-156

Table 1: The first six genus-zero GW coefficients and BPS numbers for $\mathcal{O}(1) \oplus \mathcal{O}(-3) \rightarrow \mathbb{P}^1$.

256 *4.5. Asymptotics*

257 Stirling's formula yields

$$\binom{4d}{d} \sim \sqrt{\frac{2}{3\pi d}} \left(\frac{256}{27} \right)^d,$$

258 hence

$$N_d \sim (-1)^{d+1} \frac{1}{4} \sqrt{\frac{2}{3\pi}} \left(\frac{256}{27} \right)^d d^{-7/2}.$$

259 Thus the radius of convergence of F_0 is $27/256$.

260 *4.6. General degree recursion and asymptotics*

261 For $R = (k + 1)^2$ set

$$C_d^{(k)} = d^3 N_d^{(k)} = (-1)^{(d-1)k} \binom{Rd - 1}{d - 1}.$$

262 Then

$$\frac{C_{d+1}^{(k)}}{C_d^{(k)}} = (-1)^k \frac{\prod_{j=0}^{R-1} (Rd + j)}{(d + 1) \prod_{j=0}^{R-2} ((R - 1)d + j)}.$$

263 Thus the genus-zero sequence is hypergeometric in the degree.

264 The exponential scale is

$$\rho_k = \frac{(R - 1)^{R-1}}{R^R}.$$

265 Indeed,

$$\binom{Rd}{d} \sim \sqrt{\frac{R}{2\pi(R - 1)d}} \rho_k^{-d}.$$

266 Consequently $N_d^{(k)}$ has the universal power correction $d^{-7/2}$, up to an explicit
267 k -dependent constant and parity sign.

268 *4.7. BPS asymptotics and exponentially small Möbius corrections*

269 The preceding asymptotic formula for the Gromov–Witten coefficients
270 also controls the BPS sequence itself.

271 **Theorem 4.3** (BPS asymptotics). *Let $R = (k + 1)^2$ and*

$$\rho_k = \frac{(R - 1)^{R-1}}{R^R}.$$

272 *Then, as $d \rightarrow \infty$,*

$$n_d^{(k)} \sim N_d^{(k)} \sim (-1)^{(d-1)k} \frac{1}{\sqrt{2\pi R(R - 1)}} \rho_k^{-d} d^{-7/2}.$$

273 *More precisely, for every $\eta > 0$,*

$$n_d^{(k)} - N_d^{(k)} = O_{k,\eta} \left(\rho_k^{-d/2} d^{-7/2+\eta} \right).$$

274 *In particular,*

$$\frac{n_d^{(k)}}{N_d^{(k)}} = 1 + O_{k,\eta} \left(\rho_k^{d/2} d^\eta \right).$$

275 *Proof.* By Möbius inversion,

$$n_d^{(k)} - N_d^{(k)} = \frac{1}{d^3} \sum_{\substack{m|d \\ m \geq 2}} \mu(m) A_k(d/m).$$

276 Since

$$|A_k(j)| = \frac{1}{R} \binom{Rj}{j} = O_k(\rho_k^{-j} j^{-1/2}),$$

277 we must control this bound uniformly over proper divisors. Put

$$f_k(x) = \rho_k^{-x} x^{-1/2}.$$

278 Because $0 < \rho_k < 1$,

$$\frac{d}{dx} \log f_k(x) = \log(\rho_k^{-1}) - \frac{1}{2x} > 0$$

279 for all $x \geq J_k$, where J_k is a fixed constant depending only on k . Hence f_k is
280 increasing on $[J_k, \infty)$. For a proper-divisor term $j = d/m \leq d/2$ with $j \geq J_k$,

$$\rho_k^{-j} j^{-1/2} \leq \rho_k^{-d/2} (d/2)^{-1/2}.$$

281 The finitely many values $1 \leq j < J_k$ are bounded independently of d ; since
282 $\rho_k^{-d/2} d^{-1/2} \rightarrow \infty$, they are absorbed by the same bound for all sufficiently
283 large d . Therefore, uniformly over all proper divisors,

$$|A_k(d/m)| = O_k(\rho_k^{-d/2} d^{-1/2}).$$

284 It follows that

$$|n_d^{(k)} - N_d^{(k)}| \leq \frac{\tau(d)}{d^3} O_k(\rho_k^{-d/2} d^{-1/2}).$$

285 Using $\tau(d) = O_\eta(d^\eta)$ gives the asserted error term.

286 Finally Stirling's formula gives

$$\frac{1}{R} \binom{Rd}{d} \sim \frac{1}{\sqrt{2\pi R(R-1)}} \rho_k^{-d} d^{-1/2},$$

287 and division by d^3 yields the leading asymptotic for $N_d^{(k)}$. The error is
288 exponentially smaller than the leading term, so the same asymptotic holds
289 for $n_d^{(k)}$. $\square$

290 *Remark 4.4.* Theorem 4.3 shows that the multiple-cover transform changes the
291 arithmetic nature of the coefficients without changing their leading exponential
292 geometry. Proper divisors contribute only at the secondary scale $\rho_k^{-d/2}$ or
293 smaller.

5. Conclusions

The known antidiagonal local mirror-symmetry formula for $\mathcal{O}(k) \oplus \mathcal{O}(-2-k) \rightarrow \mathbb{P}^1$ admits a simple binomial reformulation. Its Möbius multiple-cover transform belongs to a larger binomial family whose cubic prime-step congruence is completely classified by congruence conditions modulo 12. Gessel's refinement of the Jacobsthal–Kazandzidis congruence proves the required cubic supercongruence uniformly at all primes; at $p = 2$ the exceptional binomial sign is exactly canceled by the parity character dictated by the local Calabi–Yau parameter. This yields a direct arithmetic proof of the integrality of all genus-zero BPS numbers associated with (1).

The Lagrange inversion formula further gives an explicit realization of the unsigned binomial sequence from the rational function $z/(1-z)$ with parameter $k(k+2)$. Its formal similarity with framing constructions for rational 2-functions provides arithmetic context, while the present proof of the cubic congruences remains independent of framing conventions.

Natural extensions include higher-genus local curve invariants, refined BPS structures, and analogous divisibility phenomena for multi-parameter toric Calabi–Yau geometries.

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